WIVOZA · LESSON DEBRIEF

Biology · 3

Wednesday, October 14, 2026 · 10:05 AM · 9th · 22:38 recorded

LESSON AT A GLANCE

The lesson covers cellular respiration — how cells turn glucose and oxygen into usable energy — moving from a modelled walk through the equation to small groups predicting what happens when oxygen runs short.

YOU SPOKE	STUDENTS SPOKE	SILENCE / OTHER	QUESTIONS	AVG. WAIT
34.3 %	13.5 %	52.2 %	7	3.88 s

Coverage: 22:38 recorded of 22:38. A dash means Wivoza could not measure that reliably — never that the answer was zero.

1 Talk & Participation

Who was heard, and for how long.



You 34% · Students 14% · Silence 52.2%

YOUR PACE

214 wpm

Words a minute of your own speaking time. Everyday conversation runs about 150 — an anchor to place yourself against, not a target.

Nothing was transcribed until 4:30, so that much of the silence above is the recorder running before the lesson got going rather than quiet in the room.

2 Questioning & Checking

What you asked, how you checked, and how you responded.

WHAT YOU ASKED

QUESTIONS ASKED	HIGHER-ORDER	CHECKS FOR UNDERSTANDING
7	4 of 7	1

HOW YOU RESPONDED

Limited by how much of the room the microphone reached — a quiet answer leaves nothing to measure.

AVG. WAIT TIME	PAUSE AFTER AN ANSWER	FOLLOW-UPS	FEEDBACK THAT WAS SPECIFIC
3.88 s	2s	3	1 of 3

FEEDBACK MOMENTS
3

WHY WAIT TIME MATTERS

Three to five seconds of silence after a question is the single cheapest change most teachers can make. It feels much longer than it is.

3 Clarity & Content

What the lesson said it was about, in its own words.

WHAT THIS LESSON COVERED

Cellular respiration: where a cell's energy comes from, through a modelled walk through the equation and a small-group prediction task.

KEY VOCABULARY MOMENTS

"Aerobic just means it needs oxygen — that's the whole word."

4:00

CONNECTIONS TO THE WIDER WORLD

“Think about how your legs burn at the end of a sprint in gym.”
7:00

4 Climate & Routines

Counts, not scores. There is no such thing as a correct number here.

TRANSITIONS	CLEAR DIRECTIONS	STUDENT NAMES USED	REDIRECTIONS
4	6	6	3

DID YOUR ASKS HOLD?

Around 13:14 you asked for the room several times in quick succession, rising from naming a consequence to separating a student from the group.

“You two, take your packets to the back table, please.”

Asking again that soon means the first ask didn’t land.

At 9:47 your own words noted you’d already asked once — “I asked once already — eyes up here, please.”

Read from your speech alone. Whether the room settled is not something a room microphone can hear.

CORRECTION AND ENCOURAGEMENT

3 redirections · 2 encouragement detected · 1 corrective

More redirections came through than encouragement — 3 to 2.

These are read from the transcript rather than matched against a word list, so phrasing like "you did a beautiful job there" is caught. It can still miss, and anything the microphone did not pick up cannot be read at all — so treat a zero as "not found in what was captured", not as "not given".

About 48% of this recording came through as speech, so these counts describe that portion rather than the whole lesson.

5 A strength to keep

Something that worked in this lesson — keep doing it.

Your average wait time was 3.88s

Giving students real time to think before answering leads to deeper, more complete responses.

6 Notes from a subject specialist

How this topic was taught, read as a specialist in biology would.

WHAT WORKED

Building the equation up one molecule at a time — glucose in, oxygen in, then asking where the carbon goes — makes reactants and products something students track rather than copy down.

> *"So we've got glucose coming in, we've got oxygen coming in. Where does the carbon end up? Somebody tell me where the carbon goes."* (5:19)

WHAT WORKED

Naming all three stages — glycolysis, the Krebs cycle, the electron transport chain — right before the group task gives students the whole map they are predicting across.

> *"Three stages: glycolysis, the Krebs cycle, the electron transport chain. Your group is going to decide which one stops first."* (7:34)

WHERE IT COULD GO FURTHER

The sprint example is a strong bridge to fermentation — next time you could have groups predict what a muscle cell does when oxygen runs short before you name lactic acid, so the idea comes from their reasoning rather than your explanation.

> *"When you sprint, your muscles can't get oxygen fast enough. So what does the cell do? Hold that thought, because that's what your groups are figuring out."* (9:05)

WHERE IT COULD GO FURTHER

Asking students to define "aerobic" is a good vocabulary check — you could extend it by asking why the electron transport chain is the step that needs oxygen, since that is where most students' picture of respiration breaks down.

> *"Before you start, on the back of the sheet, define aerobic in your own words. You've got five minutes for the first part."* (13:00)

WHERE IT COULD GO FURTHER

When yeast comes up, tying it back to the bread-dough question from earlier would let the class see fermentation in a second organism, so it doesn't feel like a one-off fact about muscles.

> *"And yeast does this too — same idea, no oxygen. We'll come back to the bread."* (5:43)

7 Try next time

What this report suggests, from every section, in one list.

MOMENT TO REVISIT

Ask it, give pairs thirty seconds to write a one-line guess, and take two answers before you explain.

TALK & PARTICIPATION

Before explaining the group task, ask pairs to predict which stage stops first without oxygen, then build your explanation around their answers.

QUESTIONING & CHECKING

After the yeast question, add one follow-up — “why does that make the dough rise?” — before moving on.

TAKE FURTHER · CLARITY & CONTENT

Take this further: The sprint example is a strong bridge to fermentation — next time you could have groups predict what a muscle cell does when oxygen runs short before you name lactic acid, so the idea comes from their reasoning rather than your explanation.

KEEP DOING · CLARITY & CONTENT

Keep doing this: Building the equation up one molecule at a time — glucose in, oxygen in, then asking where the carbon goes — makes reactants and products something students track rather than copy down.

8 My Reflection

Your own words — the part that turns a report into a change.

STRENGTHS

The sprint example landed — groups were arguing about muscle cells before I had said the word fermentation.

GROWTH AREAS

I took the first right answer and moved on. Most answers got no second question.

MY NEXT STEP

After a student answers, ask one more “why?” before I move on.